

Sepehr Mahmoudian

SENIOR AI ENGINEER

Machine Learning • Knowledge Graphs • Multimodal & Embodied AI

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GITHUB github.com/sepahead

LINKEDIN linkedin.com/in/sepehr-mahmoudian

Berlin-Brandenburg • available from Nov 2026 • German citizen • EN fluent / DE good / Persian native



PROFILE

Builds custom agent harnesses, knowledge graphs and Graph RAG; evaluates LLMs, VLMs, Vision-Language-Action (VLA) systems and world models.

LLMs / VLMs • Knowledge Graphs • Agentic Vision • Multimodal AI • Scene Reconstruction
CONTEXT ENGINEERING • DATA PROVENANCE • MODEL VALIDATION • REINFORCEMENT LEARNING

AI ENGINEERING

10+ YEARS

RESEARCH TO PRODUCTION

PYTHON • RUST • AI / ML

PROFESSIONAL EXPERIENCE

Independent AI/ML Engineer & PhD Candidate

2020.04 - Present

Self-employed - PhD candidate at Technische Universität Darmstadt

Berlin & Darmstadt, Germany

- Building biophysical/functional neural models, PID (Partial Information Decomposition), VLA evaluation and neural control
- Building custom AI agent harnesses for domain-specific applications; multimodal 3D perception and auditable visualization

Senior AI Engineer

2026.01 - 2026.03

Neural Frames UG

Berlin & Hamburg, Germany

- Evaluated LLMs, VLMs and end-to-end pipelines, tracing runs and results with Langfuse
- Contributed to Python pipelines for model evaluation and inference

AI Engineering Resident

2025.09 - 2026.01

Merantix AI Campus (Hacker Room)

Berlin, Germany

- Built generative-media systems with Gaussian splatting and diffusion models for spatiotemporal anchoring
- Implemented confidential-computing smart contracts in Rust
- Deployed product services and data infrastructure across Cloudflare Workers, D1, R2, Google Cloud Run and AWS EC2

Senior AI Engineer

2024.07 - 2025.07

Vision Labs UG

Berlin & Hamburg, Germany

- Built workflows for prompt optimization, LLM evaluation and observability with DSPy and Langfuse
- Designed data and MLOps pipelines on Azure and Google Cloud with staging, automated tests and CI/CD

Research Associate

2021.07 - 2024.07

Freie Universität Berlin

Berlin, Germany

- Scaled and optimized large neural-network simulations in C/C++ and Python
- Built retrieval-augmented generation (RAG) workflows for technical literature and evaluated AI-agent reliability
- Applied statistical and machine-learning analysis in Python; reviewed code and research manuscripts in an international team

Research Associate

Georg-August-Universität Göttingen

Göttingen, Germany

Neural networks, Partial Information Decomposition (PID), information theory and reproducible data workflows

2019.07 - 2020.06

Research Associate

Goethe-Universität Frankfurt

Frankfurt am Main, Germany

Simulation and statistical analysis of neural-network models

2017.03 - 2019.07

Visiting Scientist

Max Planck Institute for Dynamics and Self-Organization

Göttingen, Germany

Statistical methods for neural-network analysis

2017.03 - 2019.03

Research Associate

Forschungszentrum Jülich

Jülich, Germany

Neural-network simulations and reinforcement-learning experiments

2015.02 - 2017.03

Software Developer

Centre for Content Creation

Cyberjaya, Malaysia

Award-winning iPhone/iPad applications and digital products

2010.09 - 2012.03

EDUCATION

PhD Candidate (Dr. rer. nat.)

Technische Universität Darmstadt

Darmstadt, Germany

New mathematical PID analyses and information-theoretic representation analysis; Engram’s domain-specific AI agent accelerates reproducible computational-neuroscience R&D through closed-loop neural-network simulations

2020.04 - Present

Master’s by Research in Biosciences: Neuroscience

University College London

London, United Kingdom

Computational Neuroscience • Graduated with Distinction (Top 10%, CGPA 4.0/4.0)

2012 - 2014

Bachelor of Science (Honours) in Computer Science

Anglia Ruskin University

Cambridge, United Kingdom

Graduated with First Class Honours (Top 1%, CGPA 4.0/4.0)

2009 - 2012

SELECTED AWARDS & FUNDING

EDTH / 2025

Special Mention at the European Defense Tech Hackathon Berlin 2025; co-organized with the Bundeswehr Cyber Innovation Hub

HUMAN BRAIN PROJECT

Research funding

NEAR HORIZON

Confidential-computing grant

UCL & SHEFFIELD

Competitive academic scholarships

IOS APPLICATIONS

15 international product awards



SIGNATURE

Sepehr Mahmoudian

PLACE Berlin, Germany

DATE 8 August 2026

PUBLIC ENGINEERING PORTFOLIO

GITHUB Open-source projects
github.com/sepahead

HUGGING FACE Models & datasets
huggingface.co/torusprime



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SELECTED AI ENGINEERING PROJECTS



engram

RESEARCH PROTOTYPE

CLOSED-LOOP NEURAL R&D

Domain-specific AI agent builds biophysical and functional neural models for provenance-tracked closed-loop simulation and reproducible research.

PythonKnowledge GraphsNEST



pid-rs

V0.9 PRERELEASE

INFORMATION DECOMPOSITION

Decomposes information into shared, unique and synergistic parts, with method-level provenance and explicit research boundaries.

Rust / PythonInformation TheoryStatistical ML



prisoma

0.9 CANDIDATE

AUDITABLE VLA EXPERIMENTS

Research toolkit for auditable VLA capture, intervention and replay, with canonical run logs that make claim boundaries and negative results explicit.

Rust / PythonVLA EvalsIntervention / Replay



NCP

V0.8 RELEASED

NEURAL CONTROL PROTOCOL

Project-agnostic protocol linking neural simulators and AI controllers to robots through four QoS planes, with safety-gated action and per-frame provenance.

RustCanonical JSONZenoh



manwe

ALPHA

AIRSPACE PERCEPTION

Research workbench for vision, acoustics, multi-camera geometry and tracking; model exports must pass artifact-identity and numerical gates.

PythonPyTorchModel Export



crebain

RESEARCH PROTOTYPE

SPATIAL SITUATIONAL AWARENESS

Unifies Gaussian-splat scenes, native object detection and multimodal 3D tracking in one tactical view for simulated drone operations.

RustSensor FusionNative Inference



melkor

HARDENING

GAUSSIAN-SPLAT INTEGRITY

Inspects and deterministically converts 3D Gaussian splats, surfacing numeric hazards and field provenance before pipeline use.

C++PLY / SPZAsset Validation



cobot-atlas

PUBLISHED + DOI

OPEN ROBOTICS DATASET

Citable robotics atlas of 2,023 unique GLB meshes, from industrial parts to manipulation objects, with its generation pipeline published alongside the data.

PythonHugging Face3D / VLA



galadriel

RESEARCH PROTOTYPE

CROSS-SENSOR CONSISTENCY

Fail-closed monitor for sensor disagreement on one track, using per-channel innovation tests and signed correlation over producer-declared projections.

RustNIS / CUSUMSigned Correlation



haldir

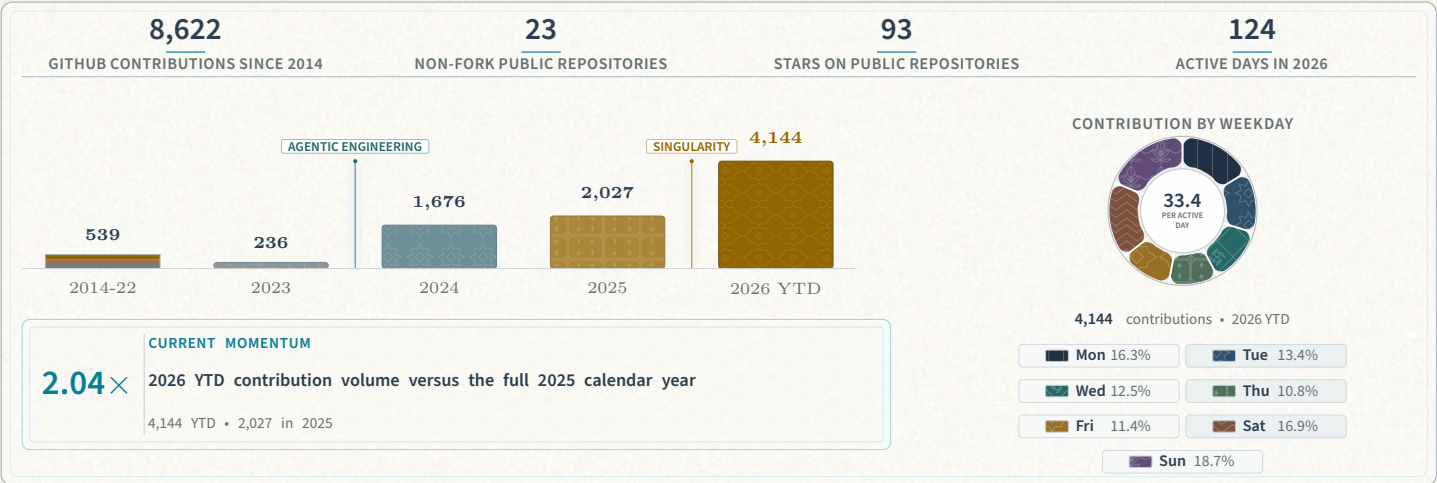
EXPERIMENTAL

MISSION AUTHORIZATION

Checks signed intents against leases, trusted state and policy; only an allow decision can construct a fresh gate-authored command.

RustAuthorizationSigned Intents

PORTFOLIO & ENGINEERING ACTIVITY



TECHNICAL SKILLS

AI / ML ENGINEERING

• AI SYSTEMS	LLMs / VLMs Agent Harnesses AI Agents Tool Use Model Evaluation Inference Pipelines
• MACHINE LEARNING	Python PyTorch Computer Vision Multimodal AI VLA Evaluation Deep Learning
• KNOWLEDGE & RETRIEVAL	Knowledge Graphs Graph RAG RAG Embeddings Vector Search Reranking Data Provenance
• SYSTEMS & MLOPS	Rust C++ FastAPI ONNX TensorRT CUDA Metal / Core ML Docker
• CLOUD & INFRASTRUCTURE	GCP AWS Azure Kubernetes Terraform GitHub Actions Linux Zenoh